

Predicting Probability of Default

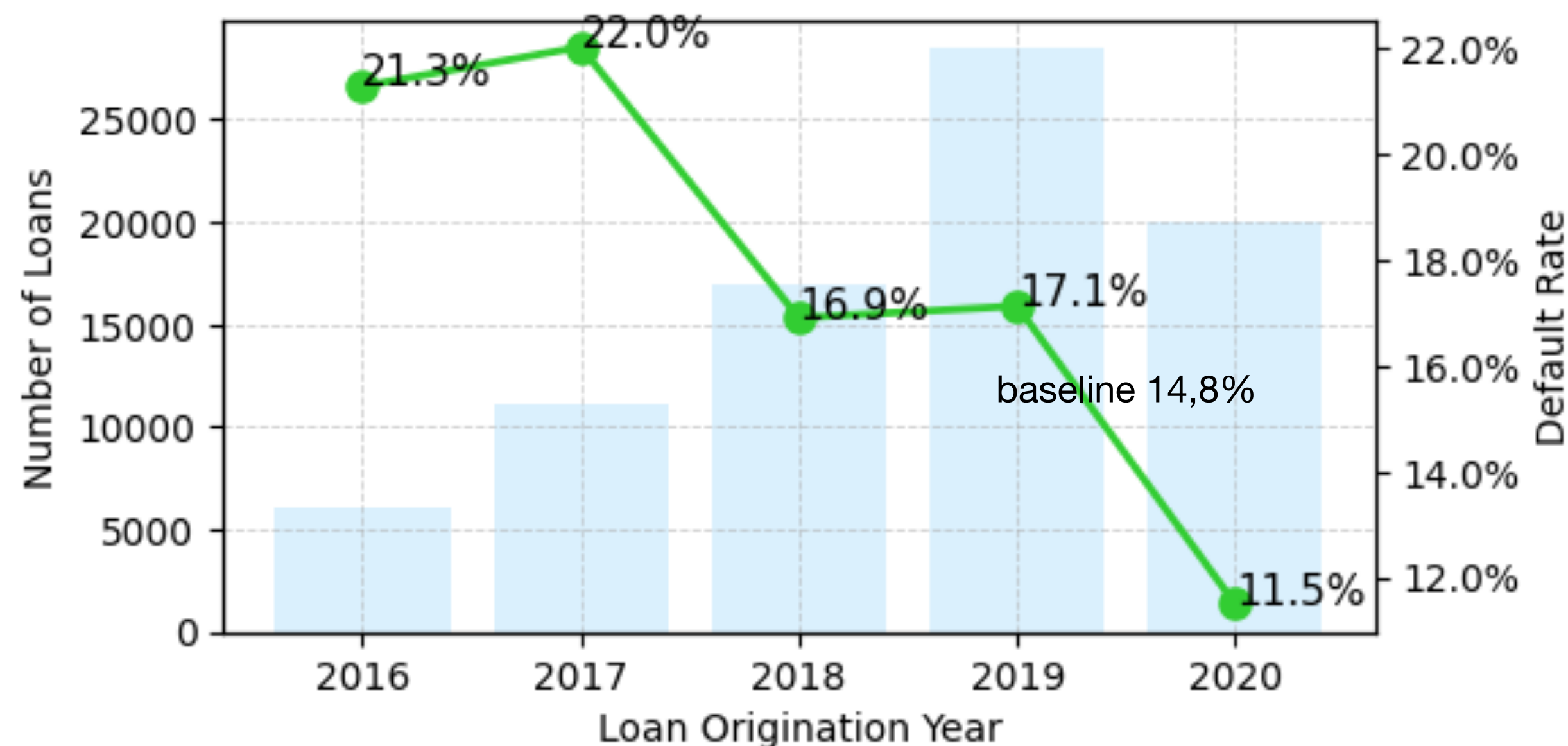
Application-time credit risk modelling on consumer loans

Problem & Dataset

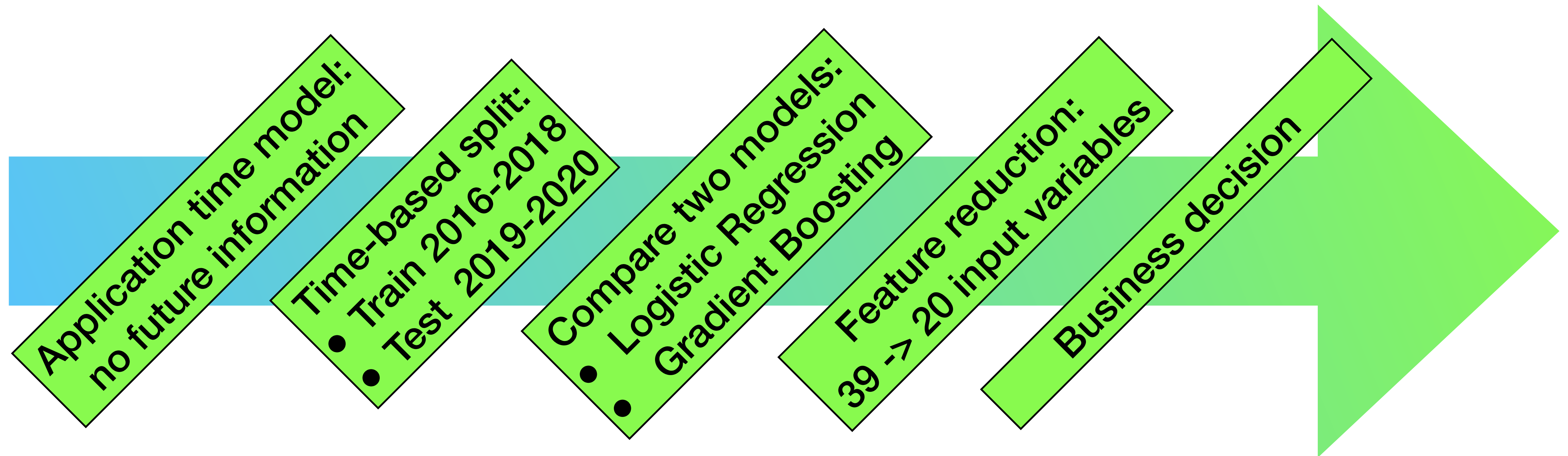
Objective: To predict the Probability of Default (PD) within 12 months for consumer loans.

Dataset: 83k historical loans from 2016 to 2020.

Ranking borrowers by risk to support lending decisions that reduce the portfolio default rate relative to the baseline (14,8%).



Modelling Approach



Logistic Regression Selected for Stability and Interpretability

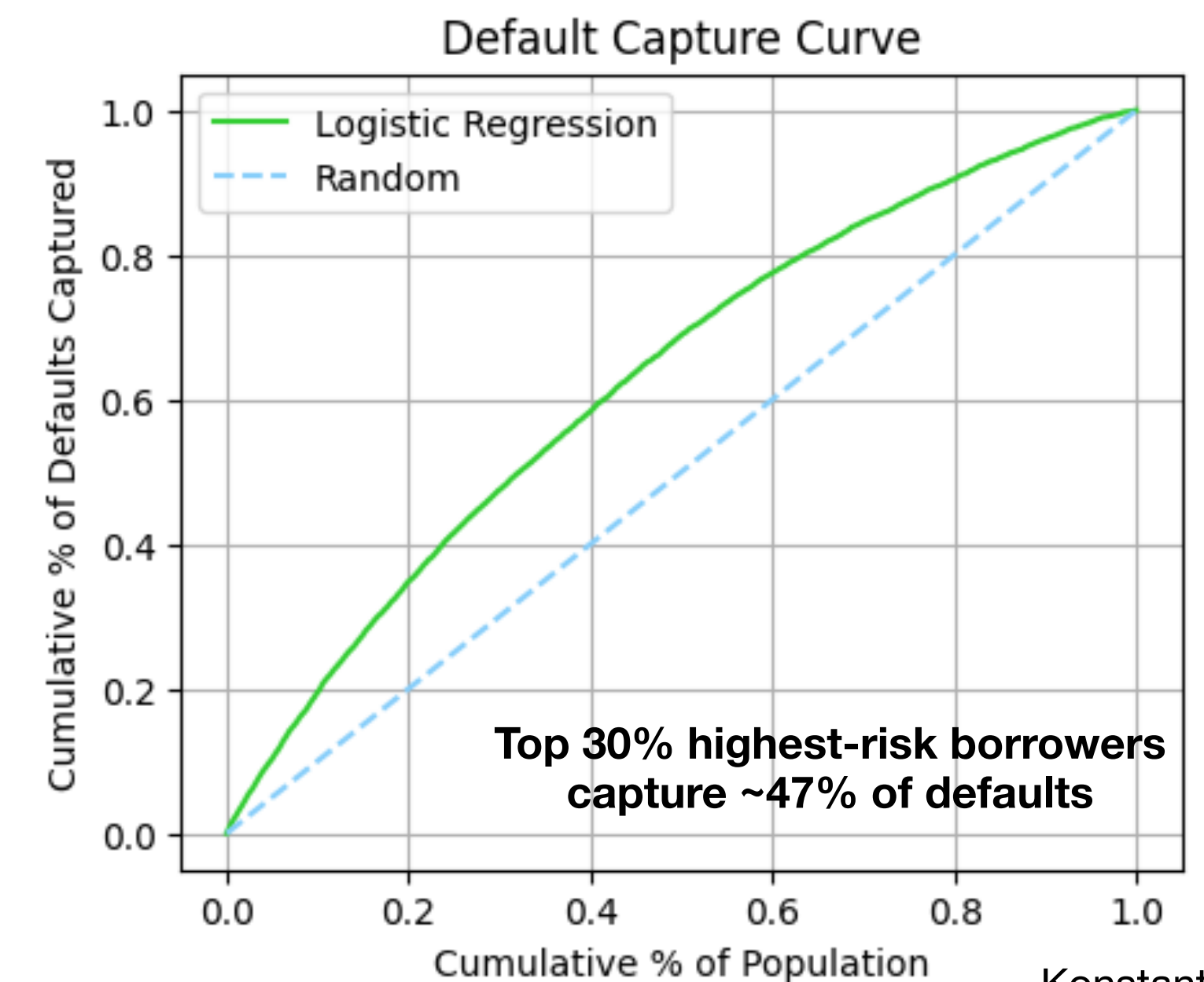
Gradient Boosting performs slightly better, but Logistic Regression generalises better and is easily interpretable.

Logistic regression with 20 features was selected due to its interpretability and stable performance.

Main risk drivers:

- ▲ New credit customer
- ▼ External credit score
- ▼ Type: mortgage
- ▼ Age

	Logistic Regression	Gradient Boosting
Number of features	20	20
TRAIN AUC	0,653	0,683
TRAIN Gini	0,306	0,367
TEST AUC	0,652	0,669
TEST Gini	0,304	0,337
Δ Gini	0,003	0,029



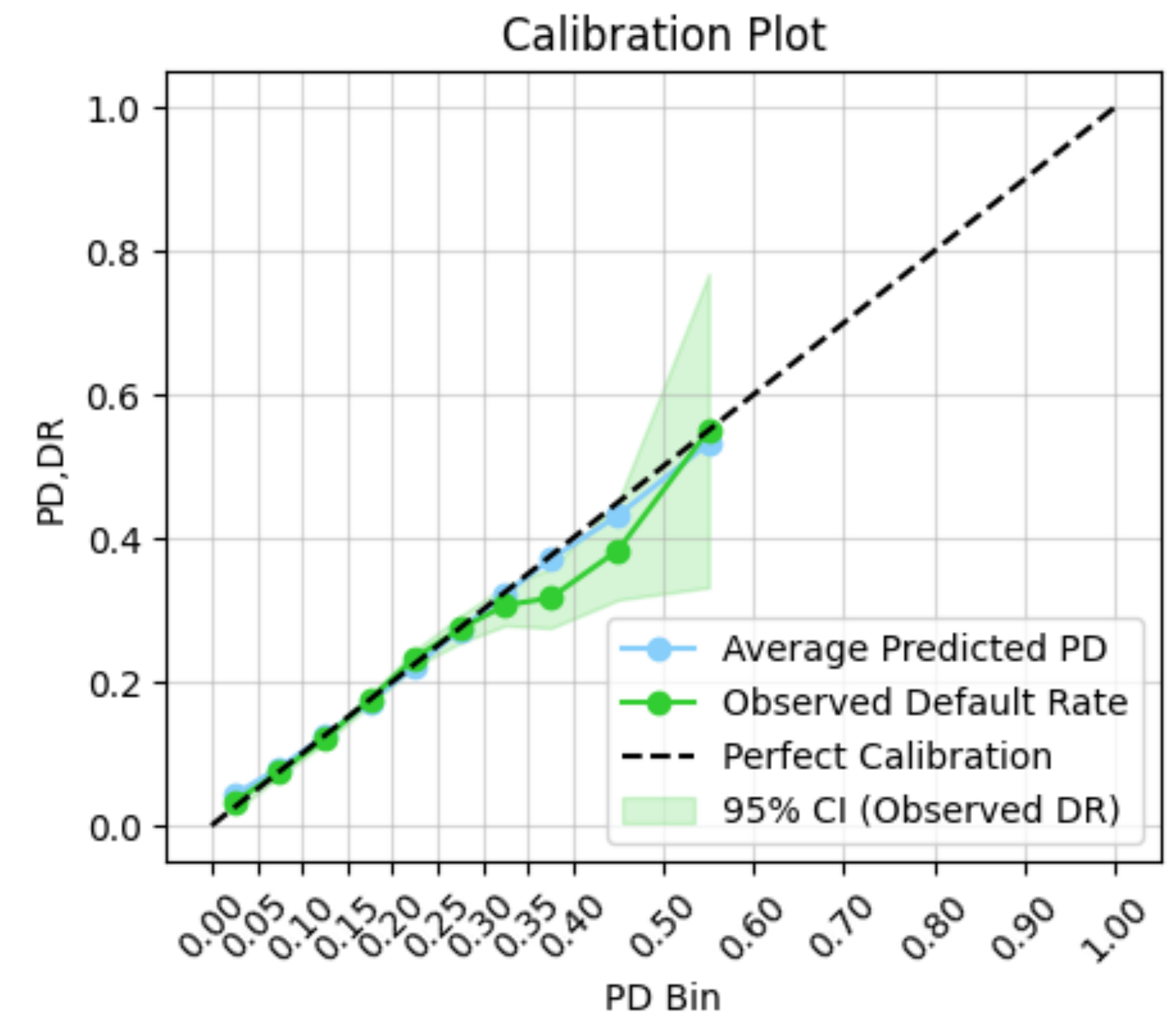
Strong Ranking and Well-Calibrated Risk in Core Portfolio

On the test period (2019–2020), observed default rates increase monotonically across PD bins, confirming strong ranking performance.

After calibration, predicted PDs closely match observed default rates across the core portfolio ($PD \leq 30\%$), with most estimates falling within confidence intervals.

At higher PD levels, some overestimation remains, but this segment represents a small share of the portfolio and exhibits high statistical uncertainty.

Overall, the model provides reliable risk estimates where it matters most, with conservative behaviour in the tail.



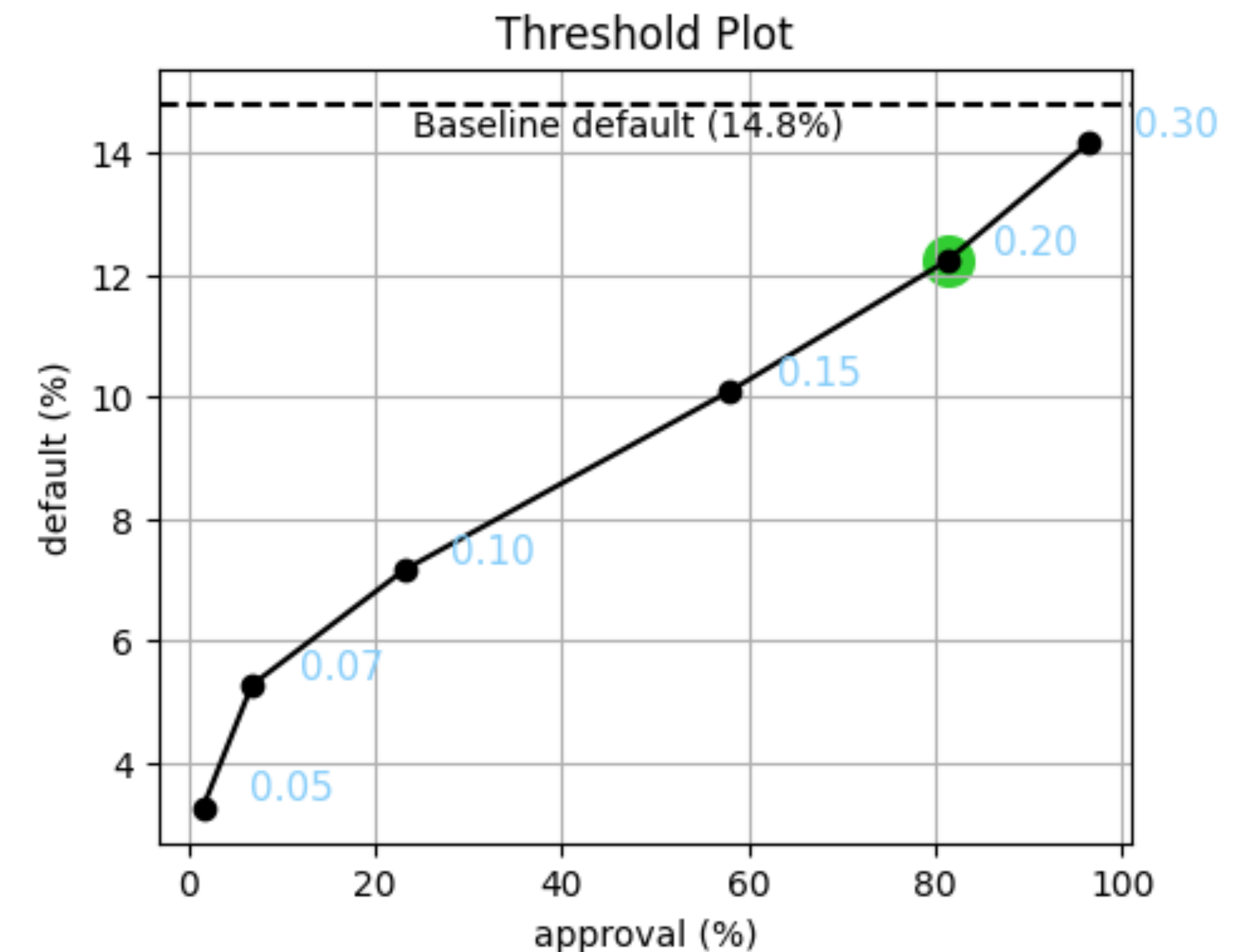
Acceptance Threshold Directly Controls Portfolio Risk and Growth

Lower thresholds reduce defaults but limit approvals.

Higher thresholds increase both approvals and defaults.

Using a 0.20 cut-off approves 81% of loans with a default rate (12.2%) lower than the 2019-2020 baseline default rate (14.8%).

Optimal threshold depends on business objective (growth vs risk vs profitability).



Deployment Considerations

Performance aligns with typical benchmarks for application-time models, where discriminatory power is constrained.

Ongoing monitoring (of acceptance, default, calibration drift) is required to ensure model stability and portfolio performance, after deployment.

Limitations include:

- ‘selection bias’ (only accepted loans are considered),
- absence of ‘reject inference’ (non-approved cases are not tracked).

The model supports decision-making but should be complemented with pricing and portfolio strategy.